

Discrete Mathematics Assignment 1

Due Date: Feb 11, 2026

Write clear and concise handwritten solutions. You must also upload a soft copy of it on the moodle page. It is fine to discuss with others, but your solutions must be in your own words that you have fully understood.

1. Consider any set S of disjoint 8's on the plane, where there is no (upper or lower) bound on the size of each 8 (nor on the shapes of the two simply closed curves that describe an 8). Two 8's are *disjoint* on the plane if the curves describing them do not intersect (for instance, one 8 placed entirely inside a closed curve describing part of another 8 is valid). What is the maximum cardinality of an arbitrary such set of 8's? Give a formal proof to justify your answer.
2. Suppose A is a countable subset of the complex numbers. Show that there exists a complex number r such that $r + A = \{r + a \mid a \in A\}$ and A are disjoint. Can you determine the cardinality of the set $D = \{r \in \mathbb{C} \mid A \cap A + r = \emptyset\}$?
3. Let D be a countable subset of $\mathbb{R}$. Is there an uncountable subset A of $\mathbb{R}^2$ such that the euclidean distance between any pair of points in A is a number in D ? Justify answer with a proof.
4. Using Zorn's lemma prove that every set can be well-ordered.
5. Let $\alpha \leq \beta$ be two infinite cardinals and let $|B| = \beta$. An α -covering of B is a collection of pairwise disjoint sets $\{A_i \mid i \in I\}$ such that $|A_i| = \alpha$ for each $i \in I$ and $B = \bigcup_{i \in I} A_i$. Does B have an α -covering? Justify answer with proof.
6. What is the cardinality of the set of all continuous functions from $\mathbb{R} \rightarrow \mathbb{R}$? Justify with a complete proof from first principles.
7. Suppose $X = (V, E)$ is a graph of maximum degree $\Delta \in \mathbb{N}$. If $|V|$ is finite then what is an upper bound on the chromatic number of X ? Is this upper bound tight? Does the same upper bound hold if the vertex set V has arbitrary cardinality? Justify answers with formal proofs.

8. Find the number of subsets of $[n]$ of size congruent to $m \pmod{3}$ (do it for each choice of $m \pmod{3}$ and $n \pmod{6}$ and tabulate the answers).